

Figure 1: A plot showing the evolution of the effective number of degrees of freedom, N_{eff} , as a function of the logarithm of the scale factor, $\ln(a)$. The x-axis ranges from $\ln(a) = 0$ to 10 . The y-axis ranges from $N_{\text{eff}} = 1.8$ to 3.0 . The plot shows a series of data points with error bars, representing the evolution of N_{eff} . A horizontal line at $N_{\text{eff}} = 2.8$ is labeled 'Statistical uncertainty'. A horizontal line at $N_{\text{eff}} = 2.6$ is labeled 'Systematic uncertainty (R16/R24)'. The data points are labeled with ' $\ln(a)$ ' values: $\ln(a) = 0$, $\ln(a) = 1$, $\ln(a) = 2$, $\ln(a) = 3$, $\ln(a) = 4$, $\ln(a) = 5$, $\ln(a) = 6$, $\ln(a) = 7$, $\ln(a) = 8$, $\ln(a) = 9$, $\ln(a) = 10$. The N_{eff} values decrease from approximately 2.7 at $\ln(a) = 0$ to approximately 1.9 at $\ln(a) = 10$.

Figure 1 is a plot showing the ratio of the number of detected events to the number of generated events for different annealing phases. The y-axis is labeled 'R' and ranges from 0.8 to 2.0. The x-axis lists annealing phases: bef_ann, ann_75C_5h, ann_75C_25h, ann_100C_5h, ann_100C_25h, ann_125C_5h, ann_125C_25h, ann_150C_5h, and ann_150C_25h. Data points are shown as black dots with vertical error bars. A horizontal line at R=1.0 is labeled 'Statistical uncertainty'. A horizontal line at R=1.7 is labeled 'Systematic uncertainty (R16/R24)'.

Figure 3 is a plot showing the relative error of the estimated parameters for the different annealing phases. The y-axis represents the relative error, ranging from 1.4 to 2.6. The x-axis lists the annealing phases: bef_ann, ann_75C_5h, ann_75C_25h, ann_100C_5h, ann_100C_25h, ann_125C_5h, ann_125C_25h, ann_150C_5h, and ann_150C_25h. For each phase, a vertical line indicates the statistical uncertainty (black error bars) and a horizontal line indicates the systematic uncertainty (purple line). The statistical uncertainty is generally larger than the systematic uncertainty, especially for the bef_ann phase.

Figure 10 is a plot showing the evolution of the 1-sigma uncertainty on the annihilation cross-section σ_{ann} as a function of the annihilation phase ϕ_{ann} . The y-axis represents σ_{ann} in units of $\text{cm}^3 \text{s}^{-1}$, ranging from 0 to 2.5. The x-axis shows the annihilation phase ϕ_{ann} from 0 to 2π . Data points are shown for different annihilation phases: $\phi_{\text{ann}} = 0$, $\pi/4$, $\pi/2$, $3\pi/4$, π , $5\pi/4$, $3\pi/2$, $7\pi/4$, and 2π . The uncertainty is highest at $\phi_{\text{ann}} = 0$ and decreases as ϕ_{ann} increases. The plot includes a legend for 'Statistical uncertainty' (black error bars) and 'Systematic uncertainty (R16/R24)' (pink error bars). A horizontal blue line is drawn at $\sigma_{\text{ann}} = 0$.

Figure 10 is a plot showing the relative uncertainty of the R16/R24 ratio for different annealing phases. The y-axis represents the relative uncertainty, ranging from 1.5 to 4.0. The x-axis lists various annealing phases: bef_ann, ann_75C_5h, ann_75C_25h, ann_100C_5h, ann_100C_25h, ann_125C_5h, ann_125C_25h, ann_150C_5h, and ann_150C_25h. Two data series are shown: 'Statistical uncertainty' (black error bars) and 'Systematic uncertainty (R16/R24)' (pink error bars). The statistical uncertainty is generally higher than the systematic uncertainty, especially for the 'bef_ann' phase.

Annealing Phase	Statistical Uncertainty (Relative)	Systematic Uncertainty (R16/R24) (Relative)
bef_ann	~3.2	~1.9
ann_75C_5h	~2.7	~2.1
ann_75C_25h	~3.0	~2.4
ann_100C_5h	~2.2	~2.2
ann_100C_25h	~2.2	~2.2
ann_125C_5h	~2.2	~2.2
ann_125C_25h	~2.2	~2.1
ann_150C_5h	~2.2	~2.2
ann_150C_25h	~2.2	~2.2